Sharp overstatement of coalition savings in newsvendor relaxations

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Abstract

Fractional coalition optimization gives an upper bound on the largest saving available to a coalition under a fixed cost allocation. We characterize its worst-case overstatement in an exclusion hierarchy for a newsvendor cost game. For every prescribed positive share vector, the supremum of the gap divided by expected grand-offer cost is the sum of the largest retained shares at each nonterminal exclusion depth. A common sequence of bounded two-state instances approaches these values, has a correct forecast of total output, and leaves every actual proper coalition strictly stable. The construction also gives a sharp root bound and a rank-one norm-cost representation. A standalone-cost condition reduces the bound; an iid validation argument and a descriptive comparison delimit its practical scope. The results concern the error of fixed-allocation fractional certification. They do not establish general statistical or computational superiority.

1 Introduction and relation to prior work

Suppose a group implements a common newsvendor decision and divides its expected cost according to fixed shares. A coalition can compare its assigned charge with the cost of acting independently. Assessing the largest saving requires a search over subsets, and allowing fractional memberships provides a continuous upper bound. A positive fractional value can nevertheless be a false indication of a profitable departure: the optimizing membership may not describe any actual coalition.

This note asks a quantitative question. For a fixed positive cost split, how large can that overstatement be after requiring several participants to be completely absent? The answer is an exact worst-case constant, indexed by the exclusion depth and the shares. The upper estimate follows from the charge budget. The matching construction realizes that estimate asymptotically while retaining positive producer variances, bounded outputs, an aggregate-correct forecast and strict stability of every proper coalition. The same sequence works at all nonterminal depths.

Fractional participation and comparisons between ordinary cores and dual allocations are established ideas. Aubin (1981) develops cooperative fuzzy games; Samet and Zemel (1984) studies LP-game cores, dual sets and replication. Denault (2001) connects homogeneous fractional cost inequalities to subdifferentials. These results motivate the distinction between integer and fractional stability, rather than making that distinction a contribution of this note.

The closest optimization formulation is the fixed-allocation subproblem in the 2005 manuscript of Burer and Dror (2005, Sections 5.2–5.3, Eq. 19). Section 4 gives its exact correspondence with our root problem for norm-valued costs. The manuscript subsequently optimizes the allocation; our result keeps it fixed. That relaxation should be cited to the 2005 manuscript: the journal version (Burer and Dror, 2012) instead develops fair allocations under optimized Gaussian correlations and does not retain the same Section 5.

Partially continuous coalition searches also appear in Bonnans and André (2008, Sections 4–6). Their selected continuous classes differ from the union of zero-coordinate faces used here.

The absolute-sum two-state cost in our example is a familiar device: Chen and Zhang (2009, Example 1 and Theorem 7) distinguish core and dual allocations and establish hardness of core membership testing. We do not claim a new hardness result, general relaxation scheme or two-state representation. The proposed increment is the exact extremal value for every prescribed positive split and exclusion depth under the stated realizability restrictions.

2 Model and exclusion hierarchy

Let $N = \{1, \dots, n\}$, $n \geq 3$, and let $Y = (Y_1, \dots, Y_n)$ be jointly distributed with $0 \leq Y_j \leq R_j < \infty$. Within-vector dependence is unrestricted. For positive penalties α, β, use

$$\ell(q, t) = \alpha(t - q)_+ + \beta(q - t)_+. \tag{1}$$

For memberships $z \in \mathbb{R}^n_+$, define the *physical fractional cost*

$$c(z) = \min_{q \in \mathbb{R}} \mathbb{E}\ell(q, z^\top Y). \tag{2}$$

An optimizer lies in $[0, R^\top z]$. Fix an implemented offer $q_0 \in [0, \sum_j R_j]$ and positive shares w_j summing to one. Set

$$L = \mathbb{E}\ell(q_0, \mathbf{1}^\top Y), \qquad a_j = Lw_j, \qquad f(z) = a^\top z - c(z).$$

Unless stated otherwise $L > 0$. The choice q_0 need not be optimal under the true law. Charges are frozen before the coalition search; they are not reallocated after observing a candidate departure.

Write $a(S) = \sum_{j \in S} a_j$ and $\mathcal{S} = \{S : \emptyset \neq S \subsetneq N\}$. The largest positive saving is

$$B = \max\{0, \max_{S \in \mathcal{S}} f(\mathbf{1}_S)\}. \tag{3}$$

We call a proper coalition strictly stable if $f(\mathbf{1}_S) < 0$. If $L = c(\mathbf{1})$, the additional efficiency condition holds and $B = 0$ means that a is an ordinary-core allocation. Without this equality, we use the excess interpretation and do not assert ordinary-core membership.

Lemma 1. *The function c is nonnegative, convex, positively homogeneous and subadditive on $\mathbb{R}^n_+$, with $c(0) = 0$.*

Proof. Nonnegativity is immediate. Scaling both q and z in (1) proves positive homogeneity. Given two optimal offers, their convex combination is an admissible offer at the corresponding convex combination of memberships. Joint convexity of the loss therefore proves convexity of the optimized value. Convexity and homogeneity give $c(z + z') = 2c((z + z')/2) \leq c(z) + c(z')$. □

At the root, define

$$P_0 = \{z \in [0, 1]^n : \mathbf{1}^\top z \leq n - 1\}, \qquad D_0 = \max_{z \in P_0} f(z).$$

For a set J of excluded participants, let $P_J = \{z \in [0, 1]^n : z_j = 0 \ (j \in J)\}$. For $1 \leq k \leq n - 1$, put

$$D_k = \max\left\{0, \max_{\substack{S \in \mathcal{S} \\ |N \setminus S| < k}} f(\mathbf{1}_S), \max_{\substack{J \subset N \\ |J| = k}} \max_{z \in P_J} f(z)\right\}. \tag{4}$$

An empty maximum is ignored. The direct integer checks retain coalitions larger than a depth-k face; omitting them would change the problem.

Proposition 2. $D_0 \geq D_1 \geq \cdots \geq D_{n-1} = B$.

Proof. Every proper integer coalition is either checked directly or belongs to one of the faces at each depth. Every depth-$k+1$ face belongs to a depth-k face, as do the extra integer coalitions introduced at depth $k+1$. All depth-one faces belong to P_0. At depth $n-1$, each face has one free coordinate, and homogeneity makes its objective linear along that segment. An endpoint attains the maximum. Together with the direct checks these endpoints recover all proper integer coalitions and zero. □

For an explicit finite distribution, the face search is a linear program: introduce an offer and positive/negative imbalance variables for each scenario in (2). At depth k it requires $\binom{n}{k}$ face problems and $\sum_{j=1}^{k-1}\binom{n}{j}$ direct checks. This is a description of the hierarchy, not a polynomial-time claim when k grows with n.

3 Sharp error for prescribed shares

Let $s_k(w)$ be the sum of the $n-k$ largest entries of w.

Theorem 3. *Fix $n \geq 3$ and $w > 0$ with $\sum_j w_j = 1$. For $1 \leq k \leq n-2$,*

$$0 \leq D_k - B \leq [Ls_k(w) - B]_+, \qquad \sup \frac{D_k - B}{L} = s_k(w). \tag{5}$$

The supremum is over bounded instances with these shares and $L > 0$. It is unchanged if one requires unit capacities, positive individual variances, two equiprobable states, unit absolute loss, an optimal true grand offer, a forecast with the correct aggregate law that induces the prescribed shares, and strict stability of every proper coalition. At $k = n-1$ the error is zero.

Proof. Upper bound. On a face with $n-k$ free coordinates, nonnegative cost gives $f(z) \leq a^\top z \leq Ls_k(w)$. Every direct integer candidate is at most B, so $D_k \leq \max\{B, Ls_k(w)\}$. Proposition 2 supplies the lower bound and terminal exactness.

Bounded construction and forecast realization. Relabel so that w_1 and w_n are the two largest shares, with $w_1 \geq w_n$; order the middle shares increasingly. For $m \geq 2$ set $H = m + 2(n-2)$. First work in unscaled units, with two equiprobable states

$$y^+ = (m, 2, \ldots, 2, 0), \qquad y^- = (0, \ldots, 0, H-1). \tag{6}$$

Their totals are H and $H-1$. Under unit absolute loss take $q_0 = H - \frac{1}{2}$, an optimal true offer, giving $L = \frac{1}{2}$. Define a forecast by $\widetilde{Y}_j = w_j T$, where T has these same two total values. Its aggregate law is correct. With score $\nu = +1$ at $T = H$ and $\nu = -1$ at $T = H-1$, one has $\mathbb{E}\nu = 0$ and

$$\mathbb{E}[\widetilde{Y}_j \nu] = \frac{w_j}{2}, \qquad \sum_j \mathbb{E}[\widetilde{Y}_j \nu] = \frac{1}{2}.$$

Thus the forecast's optimal dual allocation, normalized by its total cost, gives exactly w. Dividing all states and offers by H puts both true and forecast outputs in $[0,1]^n$, keeps every true variance positive, and changes L to $1/(2H)$. All ratios of excess to L are unchanged.

Strict stability of actual coalitions. In unscaled units let $d = y^+ - y^- = (m, 2, \ldots, 2, -(H-1))$. The cost is

$$c(z) = \frac{1}{2}|d^\top z|. \tag{7}$$

Indeed, the minimum mean absolute deviation of two equiprobable values is half their distance. A nonempty coalition not containing n has state difference at least two. A proper coalition containing n omits positive differences of total at least two; since $d^\top \mathbf{1} = 1$, its difference is at most -1. Consequently $c(\mathbf{1}_S) \geq \frac{1}{2} = L > a(S)$ for all $S \in \mathcal{S}$. Therefore $B = 0$, with strict inequalities.

Fractional witness and sharpness. Omit the k smallest middle shares, include every other positive-difference participant fully, and give participant n membership

$$r = \frac{H-2k}{H-1} = 1 - \frac{2k-1}{H-1}.$$

For $1 \le k \le n-2$ this lies in $(0,1)$. The two weighted production totals coincide, so the membership has zero cost. Its objective divided by L is

$$s_k(w) - w_n \frac{2k-1}{H-1}. \tag{8}$$

As $m \to \infty$, this tends to $s_k(w)$. The same sequence works for every nonterminal k, proving the claimed supremum. □

For equal shares the nonterminal supremum is $(n-k)/n$. In particular, for large n a fixed number of exclusions need not give a small error relative to expected cost, even when all actual coalitions are strictly stable.

Proposition 4. *For every finite member of* (6)*, expression* (8) *equals* D_k/L*, not merely a feasible lower bound.*

Proof. Direct integer values are negative. On a face omitting n, $f(z)/L = \sum_{i<n}(w_i - d_i)z_i \le 0$, since $d_i \ge 2 > w_i$. On a face containing n, fix the other memberships and write $u = \sum_{i<n} d_i z_i$. At least one positive-difference participant is absent, hence $0 \le u \le H-2$. As a function of z_n, the normalized objective has slopes $w_n + (H-1) > 0$ and $w_n - (H-1) < 0$ on either side of $u/(H-1)$. It is therefore maximized by canceling the difference: $z_n = u/(H-1) \in [0,1]$.

After cancellation, each free positive coordinate has coefficient $w_i + w_n d_i/(H-1) > 0$ and is set to one. The best omissions minimize the sum of these coefficients. Participant 1 has a coefficient at least as large as every middle participant because $m \ge 2$ and $w_1 \ge w_i$. Among the middle participants the added coefficient is the same. Thus omitting the k smallest middle shares is optimal, giving (8). □

Remark 5 (Normalization). The sharpness denominator is L, not the worst possible loss of the implemented offer over the declared box,

$$A = \max\{\beta q_0, \alpha(\textstyle\sum_j R_j - q_0)\}.$$

In the unit-capacity version of (6), $q_0 = 1 - 1/(2H)$ and $A = n - 1 + 1/(2H)$. Hence

$$\frac{L}{A} = \frac{1}{2H(n-1)+1} \longrightarrow 0.$$

The limiting construction does not establish a large error relative to A or explain a finite-data certificate's failure at a fixed percentage-of-A precision target.

4 Root relaxation and the choice of extension

4.1 Correspondence with the earlier norm-cost formulation

Put $P_1 = \{z \in [0,1]^n : 1 \le \mathbf{1}^\top z \le n-1\}$. Positive homogeneity gives

$$D_0 = \max\{0, \max_{z \in P_1} f(z)\}. \tag{9}$$

To see this, a positive objective at a point of mass $s < 1$ increases when z is replaced by z/s. Each coordinate remains at most one because $z_j \le s$; the new mass is one. Nonpositive objectives do not improve on zero.

For $c(z) = \|Mz\|_2$, Eq. 19 of Burer and Dror (2005) has value $v(a) = \min_{z \in P_1}\{c(z) - a^\top z\}$. Thus $D_0 = [-v(a)]_+$. The analogous binary program maps to B. This identity concerns a fixed a, and does not assess the quality of an allocation selected by an outer optimization.

Corollary 6. *For the model and prescribed shares of Theorem 3,*

$$0 \le D_0 - B \le \Big[L(1 - \min_j w_j) - B\Big]_+, \qquad \sup \frac{D_0 - B}{L} = 1 - \min_j w_j.$$

Proof. $D_0 \le \max_{z \in P_0} a^\top z = L(1 - \min_j w_j)$. Since $D_0 \ge D_1$, the depth-one construction in Theorem 3 proves the matching supremum. □

The same cost obstruction belongs to the rank-one norm class: (7) is $\|Mz\|_2$ with $M = d^\top/2$. For a Gaussian realization, let Z be standard normal, $\eta = \mathbb{E}|Z| = \sqrt{2/\pi}$, and $X = \mu + dZ/(2\eta)$. The optimized absolute-loss cost of $z^\top X$ is $|d^\top z|/2$. Taking $\mu \ge 0$ also makes the optimal offer $\mu^\top z$ nonnegative. The covariance is singular and Gaussian outputs are not bounded or almost surely nonnegative. Those physical properties are supplied by the separate two-state realization, not by the Gaussian one.

4.2 Canonical and physical fractional costs

The canonical concave cover is an established construction (Samet and Zemel, 1984; Shellshear, 2009). In cost signs it reads

$$\bar{c}(z) = \min_{\lambda_S \ge 0} \left\{ \sum_{\emptyset \ne S \subseteq N} \lambda_S c(\mathbf{1}_S) : \sum_S \lambda_S \mathbf{1}_S = z \right\}. \tag{10}$$

This is not generally the physical cost (2).

Proposition 7. *For the sublinear cost c, $c(z) \le \bar{c}(z)$ and $\bar{c}(\mathbf{1}_S) = c(\mathbf{1}_S)$ for every actual coalition. If a is an ordinary-core allocation, then $a^\top z \le \bar{c}(z)$ for every nonnegative membership.*

Proof. Subadditivity and homogeneity bound $c(z)$ by every feasible objective in (10). At $z = \mathbf{1}_S$, using S alone proves equality. For a core allocation, multiply $a(S) \le c(\mathbf{1}_S)$ by λ_S and sum, then minimize. □

For example, use unscaled states $(2, 2, 0)$ and $(0, 0, 3)$ and equal charges $a_j = 1/6$. Every actual proper coalition is strictly stable. At $z = (1, 0, 2/3)$, however,

$$c(z) = 0, \qquad a^\top z = 5/18, \qquad \bar{c}(z) = 2/3.$$

For the last equality, only $\{1\}$, $\{3\}$ and $\{1, 3\}$ can be used in (10). If the pair coefficient is $t \le 2/3$, the cost is $2 - 2t$, minimized at $t = 2/3$. Thus ordinary-core preservation for a canonical extension does not imply exactness of the physical fractional audit.

5 A condition that reduces the error

Let $c_j = c(\mathbf{1}_{\{j\}})$ and define

$$\rho = \min(\{1\} \cup \{a_j/c_j : c_j > 0\}).$$

In particular, $\rho = 1$ when all singleton costs vanish.

Proposition 8. *For $1 \le k \le n - 2$,*

$$D_k - B \le (1 - \rho)\,[Ls_k(w) - B]_+. \tag{11}$$

Proof. The case $\rho = 0$ is Theorem 3. On a depth-k face with free set F put $x = \sum_{j \in F} a_j(1 - z_j)$. Nonnegative cost gives $f(z) - B \le a(F) - B - x$. Also,

$$c(\mathbf{1}_F) \le c(z) + \sum_{j \in F} (1 - z_j)c_j \le c(z) + x/\rho.$$

As F is a proper actual coalition, $B \ge a(F) - c(\mathbf{1}_F)$. It follows that $f(z) - B \le (1/\rho - 1)x$. The smaller of the two bounds is at most their weighted average with weights $1 - \rho$ and ρ, which cancels x and leaves $(1 - \rho)(a(F) - B)$. Take the maximum over faces, zero and the direct integer candidates, all of whose excess above B is nonpositive. Use $a(F) \le Ls_k(w)$. □

For example, covering at least half of every singleton's optimized cost halves the share-only ceiling. This is an elementary sufficient condition from subadditivity, not an independent priority claim. The root is not included in (11); its maximizer can have all coordinates positive. If $L = 0$, all positive excesses and all D_k are zero directly.

Another sufficient condition is symmetry. Under exchangeable production and equal charges $a_j = a_0$, fix a face F with $|F| = n - 1$. Averaging memberships over its permutations, then using convexity and homogeneity, gives $c(z) \ge (\sum_{j \in F} z_j)c(\mathbf{1}_F)/(n - 1)$. Its maximum excess is therefore attained at zero or $\mathbf{1}_F$, so $D_1 = B = [(n - 1)a_0 - c(\mathbf{1}_F)]_+$. Equal marginal laws alone do not imply this exchangeability assumption.

6 Numerical checks and scope of validation

The calculations support implementation checks, not proof or bibliographic priority. The complete deterministic grid uses $n \in \{3, 6, 8\}$, $m \in \{2, 20, 200\}$ and two share vectors: uniform shares and shares proportional to $(n, 1, 2, \ldots, n - 2, n - 1)$. All 18 laws have strictly stable actual proper coalitions. Exact enumeration and the face implementation agree with (8) at all 84 depth values, including the terminal levels. Table 1 shows the uniform six-participant slice.

m	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
2	0.814815	0.611111	0.407407	0.203704	0.000000
20	0.827160	0.648148	0.469136	0.290123	0.000000
200	0.832528	0.664251	0.495974	0.327697	0.000000
Supremum	5/6	2/3	1/2	1/3	0

Table 1: Checked values of D_k/L with $B = 0$, six uniform shares and $H = m + 8$. The last row is the proved supremum; terminal depth is exact. Values are generated from the saved deterministic results.

Two separate primal LPs also agree with the existing root dual implementation on the same 18 laws. The rank-one and physical costs agree at 1,968 checked membership points, and a separate cover LP verifies the three-participant example in Section 4. These checks do not assess a reoptimized allocation or claim coverage from a finite test count.

Appendix A turns (11) into an iid validation bound using established statistical tools. A prespecified development comparison used 96 positive-share laws and 12 exchangeable controls. Four original cases with nonpositive shares were excluded before sampling. One independent pilot and validation stream per law supplied prefixes of $M \in \{256, 1024, 4096\}$ observations each. The fixed rule was independent of both streams. Each of the 18 method–size combinations used $\delta = 0.05/18$, yielding simultaneous validity across the 18 comparisons within a law, not joint coverage across all 108 laws.

The alternatives are a Bonferroni-calibrated version of the paired ratio procedure; a marginal DKW bound using both halves for fixed-rule moments; a capacity bound using both halves; a

preceding bound from pilot-fitted dual prices on faces; and paired SAA blocks of 64 observations. The latter specialize the optimality-gap construction of Mak et al. (1999), with bounded finite-sample calibration. Each candidate-dependent witness is evaluated only on validation data. Averages and counts describe this development grid, not repeated-sampling power or a controlled runtime comparison.

Method	Mean G/A	$G \le 0.05A$
Paired ratio (IUT/fixed sequence)	0.176440	5/47
Bonferroni ratio	0.177906	4/47
Marginal DKW, pooled moments	0.170268	1/47
Capacity bound, pooled moments	0.201686	0/47
Pilot-fitted face prices	0.143603	0/47
Paired SAA blocks (size 64)	0.217701	0/47

Table 2: The 47 heterogeneous positive-share laws at $M = 4096$ pilot plus $M = 4096$ validation observations. The estimand is $D_1 - B$. Here G is its upper confidence bound and A is the maximum declared grand-offer loss. All six prespecified methods are shown.

Among the 35 original laws with true $\rho \ge 0.6$, the paired, Bonferroni and DKW procedures all certified $\rho \ge 0.5$. The broader target, a certified gap at most $0.05A$ in at least half the heterogeneous laws, failed: the paired procedure passed 5 of 47. Even exact population values in (11) meet that target in only 8 of 47. Thus improving scalar estimation alone cannot eliminate this corollary's population limitation.

Post-run combinations of the paired and face bounds yield a mean normalized upper gap of 0.128273, versus 0.129887 for DKW plus face, a descriptive difference of about 1.24%. These combinations do not establish general superiority. The observed run had no gap-coverage failures, but the probability guarantees depend on the assumptions and proof. The SAA baseline used 9,072 blocks and 50,232 LP solves across the study; its fixed block choice does not settle the bias–concentration tradeoff.

7 Discussion

The exact error constant depends only on the retained shares, but its sharpness is realized by a specialized dependence structure. The result warns against interpreting a large fractional objective as evidence of a profitable actual departure, even with an aggregate-correct forecast. Whole-participant exclusions reduce the ceiling but need not make it small relative to expected cost.

Several limits remain essential. The scale issue in Remark 5 prevents an interpretation as a fixed maximum-loss error. Correct aggregate forecasting permits individual forecast errors. The bound concerns this particular physical relaxation hierarchy and does not exclude better certificates or algorithms. The standalone condition is useful only on a restricted part of the tested grid. No guarantee for dependent time-series validation, real-data forecasting advantage or general runtime superiority follows.

The focused source comparison did not locate the exact depth/share formula among the inspected results. This is not a certificate of global priority. The intended contribution is the quantitative extremal calculation with its realizability properties; established optimization and statistical ingredients are credited explicitly. The present document is a working draft for assessment of that contribution's significance.

A A finite-data conversion of the standalone condition

Assume bounded iid validation observations independent of a pilot and of the fixed rule (q_0, w). Within-vector dependence remains arbitrary. Condition on the pilot and fit offers $q_j \in [0, R_j]$. For a predetermined threshold $t \in (0, 1]$, use paired variables

$$X_{j,t} = w_j \ell(q_0, Y(N)) - t\ell(q_j, Y_j).$$

Write $b_j = \mathbb{E}\ell(q_j, Y_j) \geq c_j$. Let $\ell_{j,t}$ be a scalar lower confidence bound satisfying

$$\Pr\{\ell_{j,t} > \mathbb{E}X_{j,t} \mid \text{pilot}\} \leq \eta. \tag{12}$$

These pointwise bounds need not hold jointly.

Lemma 9. *For a fixed increasing grid $0 < t_1 < \cdots < t_K \leq 1$, test a threshold only if all $\ell_{j,t} > 0$. Stop at the first failed threshold and let $\widehat{\rho}_{\text{grid}}$ be the last passed threshold, or zero. Then $\Pr\{\widehat{\rho}_{\text{grid}} > \rho \mid \textit{pilot}\} \leq \eta$.*

Proof. If $t > \rho$, some j has $c_j > 0$ and $a_j < tc_j \leq tb_j$. Hence $\mathbb{E}X_{j,t} < 0$, and passing the threshold implies the event in (12) for this one index. For the sequential procedure, overstatement of ρ requires passing the first fixed grid point above ρ, if any. Its failure probability is at most η. □

The reasoning is the intersection–union and fixed-sequence principle (Berger and Hsu, 1996; Hsu and Berger, 1999). It removes the need for producer/grid multiplicity factors for this single global claim. It does not give simultaneous coverage of all scalar intervals or dimension-independent power.

For completeness, the implemented scalar bounds combine Hoeffding and empirical Bernstein inequalities (Maurer and Pontil, 2009, Theorem 4). For a variable in an interval of width W, $M \geq 2$ observations and unbiased sample variance v, use

$$r_\eta = \min\left\{ W\sqrt{\frac{\log(2/\eta)}{2M}},\ \sqrt{\frac{2v\log(4/\eta)}{M}} + \frac{7W\log(4/\eta)}{3(M-1)} \right\}.$$

Each inequality receives tail $\eta/2$; a union bound validates their minimum. The lower bound is the sample mean minus r_η, clipped to the known range. Here $X_{j,t} \in [-tA_j, w_j A]$, where $A_j = \max\{\beta q_j, \alpha(R_j - q_j)\}$. For a two-sided interval, split its budget again between tails.

A pilot-selected coalition $S \in \mathcal{S}$, with offer $q_S \in [0, R(S)]$, has feasible excess

$$e_S = w(S)L - \mathbb{E}\ell(q_S, Y(S)) \leq B.$$

A scalar lower confidence bound, clipped below at zero, gives $W_- \leq B$. The empty witness instead gives $W_- = 0$ exactly. Obtain a two-sided grand-cost interval $[L_-, L_+]$ and split the total budget δ into three equal parts: the grid, the grand interval and this witness.

The deterministic bound $c_j \leq C_j = \alpha\beta R_j/(\alpha + \beta)$ follows by using the offer $\alpha R_j/(\alpha + \beta)$ and comparing the two endpoint losses. Define

$$\widehat{\rho}_{\text{cap}} = \min(\{1\} \cup \{w_j L_-/C_j : C_j > 0\}), \qquad \widehat{\rho} = \max\{\widehat{\rho}_{\text{grid}}, \widehat{\rho}_{\text{cap}}\}.$$

The grand interval is clipped to $[0, A]$.

Proposition 10. *With probability at least $1 - \delta$, simultaneously for $1 \leq k \leq n - 2$,*

$$\widehat{\rho} \leq \rho, \qquad D_k - B \leq (1 - \widehat{\rho})\,[s_k(w)L_+ - W_-]_+ .$$

The terminal-depth error is zero.

Proof. Lemma 9, the grand interval and witness bound hold together by a three-event union bound. On this event the cap bound also holds, as it uses only L_-. Apply Proposition 8 and monotonicity in ρ, L, B. The same event proves every depth bound. □

The implementation uses the grid $0.05, 0.10, \ldots, 1.00$. The pilot selects from leave-one-out coalitions and their fitted offers, or an empty witness. A numerical inconsistency triggers the safe fallback $\widehat{\rho} = 0$, $W_- = 0$, $L_+ = A$. This fallback is recorded and did not occur in the saved comparison. This appendix is a conversion of an elementary population condition using established methods.

B Reproducibility and review status

The accompanying repository contains the full proofs underlying earlier diagnostics, source-by-source reading depth, protocols, negative results and source hashes. The manuscript tables are generated from existing JSON outcomes by `paper/build_tables.py`; no new sampling is used to assemble them. The deterministic exclusion family and root correspondence are regenerated by `scripts/check_exclusion_ceiling.py` and `scripts/check_exclusion_prior_art.py`.

The source comparison distinguishes the 2005 manuscript from the later journal version and records partial reading where full originals were not obtained. Tests and this draft's internal review are not independent mathematical review. No submission or external-review outcome is claimed.

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